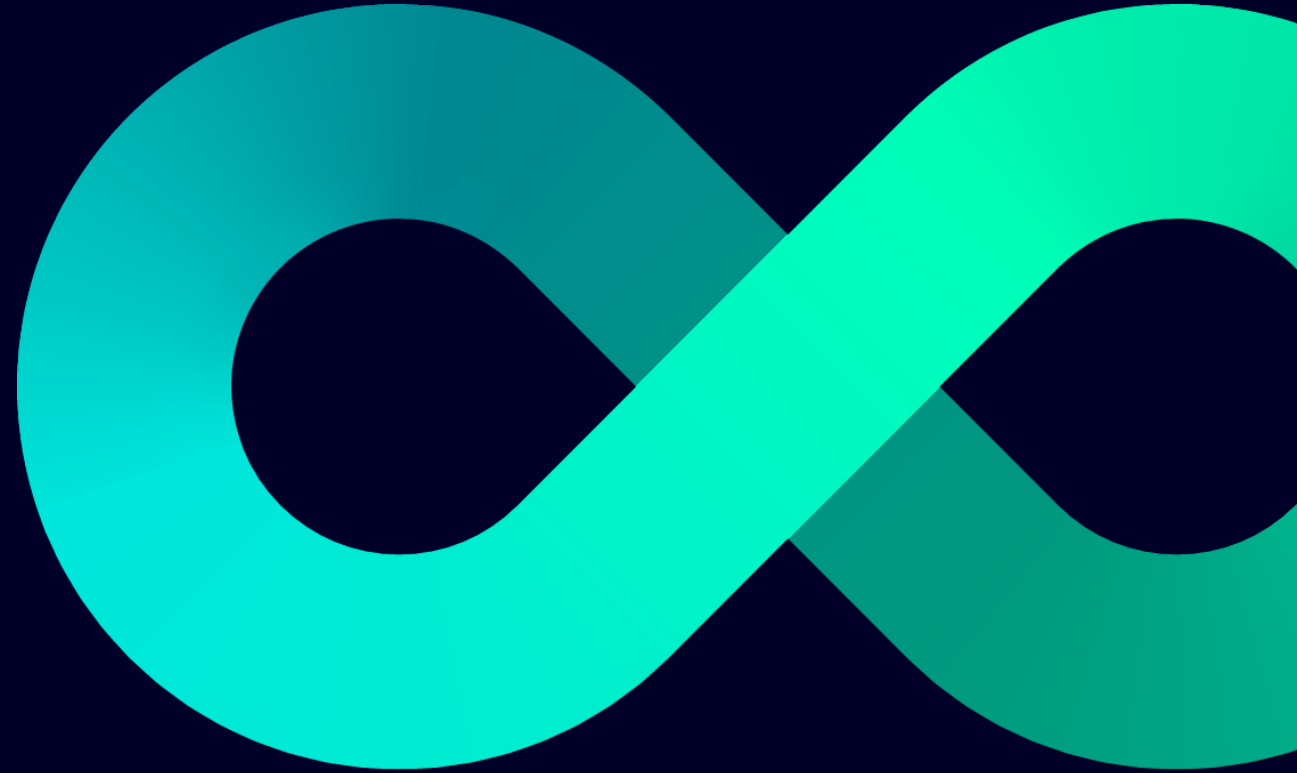


Tutorial: Hackathon Problem 1

Aug 13



Dataset

MachinePlan-10k/

```
-- part_00001/
| |-- part_00001.stp          % STEP export of CAD model
| |-- part_00001_operations.json % CAM operation sequence metadata
| |-- front_wireframe.png     % front view wireframe image of CAD model
| |-- top_wireframe.png       % top view wireframe image of CAD model
| |-- right_wireframe.png     % right view wireframe image of CAD model
| |-- isometric_shaded.png    % shaded image of CAD model
| |-- 000_BLANK_text.stl.txt   % IPW mesh before first operation
| |-- 001_NXT0302_009_details.txt % operation 1: tool/operation report
| |-- 001_NXT0302_009_text.stl.txt % operation 1: IPW mesh
| |-- 001_NXT0302_009.ptp      % operation 1: toolpath
| |-- 002_NXT0302_007_details.txt
| |-- 002_NXT0302_007_text.stl.txt
| |-- 002_NXT0302_007.ptp
| |-- ...
| |-- 011_UGT0301_282_details.txt % operation N: tool/operation report
| |-- 011_UGT0301_282_text.stl.txt % operation N: IPW mesh
| |-- 011_UGT0301_282.ptp      % operation N: toolpath
```

Different File Types

- B-Rep (.stp) [available at inference]
- CAD images (.png) [available at inference]
- Meshes (.stl/.stl.txt) [only for training, prediction target]
- Operation and tool information (details.txt) [only for training, prediction target information]
- NC-Code (.ptp) [only for training, prediction target information]

How to Work with Each File Type

- .stp, .stl.txt can be opened by any text editor
- .stp can also be opened by most CAD software
- .stl can be opened using MeshLab (open source)
- Python or other programming languages can be used to preprocess the .stp and .stl files

Output formats

- .json
- .stl
- .ptp
- See Github for full sample files

Operation Vocabulary

Main Label (O1)

- hole_making
- mill_contour
- mill_planar

Sub Label (O2)

- DRILLING
- SPOT_DRILLING
- DEEP_HOLE_DRILLING
- HOLE_MILLING
- BORING_REAMING
- AREA_MILL
- FLOOR_WALL

Tool Vocabulary

Tool type

- chamfer_mill
- twist_drill
- end_mill
- spot_drill
- insert_drill
- gun_drill
- spade_drill
- boring_tool

Please follow the given vocabulary

Helper Script to check formatting

We have provided a helper script to check if the outputs generated follow the prescribed format

- `validate_submission.py` on github

Data Usage

If you are doing any quality-based data filtering and augmentations to the dataset, please report it clearly